

HYBRIDBERT AND METADATA DEEP LEARNING MODEL FOR TWITTER BOT DETECTION

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This is to certify that the project that is entitled with the name “**HYBRIDBERT AND METADATA DEEP LEARNING MODEL FOR TWITER BOT DETECTION**” is Bonafide work done by **SHAIK SHAKEER AHAMAD (22471AO5O1)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in the Department of **COMPUTER SCIENCE AND ENGINEERING** during 2025-2026.

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PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **PO5: Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Project Course Outcomes (CO'S):

CO421.1: Analyze the System of Examinations and identify the problem.

CO421.2: Identify and classify the requirements.

CO421.3: Review the Related Literature

CO421.4: Design and Modularize the project

CO421.5: Construct, Integrate, Test and Implement the Project.

Course Outcomes – Program Outcomes mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C421.1		✓											✓		
C421.2	✓		✓		✓								✓		
C421.3				✓		✓	✓	✓					✓		
C421.4			✓			✓	✓	✓					✓	✓	
C421.5					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Course Outcomes – Program Outcome correlation

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C421.1	2	3											2		
C421.2			2		3								2		
C421.3				2		2	3	3					2		
C421.4			2			1	1	2					3	2	
C421.5					3	3	3	2	3	2	2	1	3	2	1

Note: The values in the above table represent the level of correlation between CO's and PO's:

1. Low level
2. Medium level
3. High level

Project mapping with various courses of Curriculum Attained POs:

Name of the course from which principles are applied in this project	Description of the device	Attained PO
C2204.2, C22L3.2	The first step in our project was to carefully gather and analyze the requirements to establish a clear understanding of the problem—identifying and classifying Twitter accounts as either human-operated or automated (bots). After examining different approaches, we selected a suitable process model to structure the development.	PO1, PO3
CC421.1, C2204.3, C22L3.2	Each and every requirement is critically analyzed, the process mode is identified	PO2, PO3
CC421.2, C2204.2, C22L3.3	Each module was tested individually, integrated, and evaluated using metrics such as accuracy, F1-score, and ROC-AUC.	PO3, PO5, PO9
CC421.3, C2204.3, C22L3.2	The project was organized into modules including dataset preprocessing, BERT-based text feature extraction, metadata feature engineering, and a hybrid classification layer.	PO1, PO5
CC421.4, C2204.4, C22L3.2	Documentation was jointly prepared by the team, with regular progress presentations at each stage.	PO10
CC421.5, C2204.2, C22L3.3	The Hybrid BERT + Metadata model was then trained and tested on the TwiBot-20 dataset, showing strong performance in distinguishing bots from human accounts.	PO10, PO11
C2202.2, C2203.3, C1206.3, C3204.3, C4110.2	Future improvements may include real-time Twitter API integration, ensemble learning with additional models, and deployment as a user-friendly application.	PO4, PO7

ABSTRACT

This paper proposes a deep learning-based approach for detecting Twitter bots by integrating two distinct streams of user information: tweet content and behavioral metadata. The proposed architecture combines BERT, a transformer-based language model, to capture the linguistic patterns in user tweets, alongside a parallel lightweight neural network that processes structured attributes such as follower count, posting frequency, and verification status. By fusing textual content and user behavioral features, the model effectively distinguishes between human-operated and automated (bot) accounts. The system is trained and evaluated on the TwiBot-20 dataset, which offers a comprehensive set of annotated Twitter profiles. To mitigate class imbalance during training, we employ Focal Loss, and optimization is carried out using the Adam optimizer with a low learning rate over five epochs. Our approach demonstrates strong performance, achieving over 94% accuracy and an F1-score of 0.935. The model's robustness is validated through detailed evaluation using ROC curves and confusion matrices. Furthermore, the system supports real-time prediction, making it suitable for deployment in large-scale content moderation pipelines. The model architecture is also adaptable to future enhancements, such as multilingual bot detection.

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• INTRODUCTION

Social media platforms such as Twitter are increasingly affected by automated accounts, commonly known as bots, which have become a serious threat to the authenticity and reliability of online interactions. These accounts create significant challenges because of their diverse behaviors, complex strategies, and varying levels of influence, ranging from harmless automation to harmful activities such as misinformation and manipulation. Detecting and accurately distinguishing between human users and bots is essential to preserve digital trust, ensure credible information flow, and support effective countermeasures. Traditional detection methods that rely only on text-based analysis often struggle with accuracy since many bots generate content that closely resembles human activity. This limitation highlights the importance of hybrid detection frameworks that integrate both textual patterns and user metadata to improve reliability and efficiency.

The influence of bots extends worldwide, and their role in shaping discussions has made them a subject of concern for individuals, organizations, and governments alike. While some automated accounts perform benign tasks such as news updates or weather alerts, malicious bots are designed to amplify spam, spread disinformation, manipulate public opinion, and disrupt digital communities. The rapid increase in such accounts can be attributed to the availability of automation tools, advanced generation techniques, and the lack of robust detection systems in conventional approaches. Public awareness about the presence and risks of these bots also remains limited, which often results in delayed detection and prolonged influence.

Recent research estimates suggest that automated accounts form a considerable portion of Twitter's total user base, with figures ranging between 9% and 15%. Their activities are especially concerning during sensitive events such as elections, pandemics, and social movements, where large-scale dissemination of misleading content can impact decision-making and public perception. Among the most disruptive are political bots, which operate in a coordinated manner, post at very high frequencies, and significantly influence online discussions.

A major difficulty in combating bots lies in the lack of early detection. Many users fail to differentiate between authentic and automated content, and intervention by platforms usually comes after harmful effects are already widespread. Signs of

suspicious activity often include excessive posting frequency, unusual follower-to-friend ratios, repetitive or low-quality posts, and irregular metadata patterns. Detecting these behaviors early and with precision is critical for limiting the harmful effects of malicious bots.

Another major challenge in detecting bots is the increasing sophistication of their behavior. Malicious accounts are no longer limited to simple spam generation; instead, they now replicate human-like interactions, post contextually relevant content, and exploit trending topics to avoid detection. While leading research institutions and technology companies are developing advanced detection frameworks, smaller organizations and general users often lack access to such resources. Furthermore, there is a shortage of standardized detection systems, and platform-based interventions are often delayed, allowing malicious accounts to influence discussions for extended periods.

By combining both text-based learning and structured metadata, hybrid models integrate the semantic understanding of transformers with the discriminative power of machine learning classifiers, creating a powerful solution for robust bot detection..

This research proposes an innovative hybrid model that integrates **BERT-based textual embeddings with metadata-driven Multi-Layer Perceptrons (MLP)** for Twitter bot classification. The model incorporates several advanced preprocessing strategies to ensure effective feature representation and accurate predictions. Preprocessing involves data normalization for metadata, tokenization for textual fields, and class balancing techniques to address the issue of dataset imbalance. Together, these steps prepare the input for effective fusion and analysis.

The proposed Hybrid BERT + Metadata model is designed to achieve higher accuracy and reliability in detecting bots compared to standalone approaches. While BERT ensures semantic understanding of text, the metadata network strengthens detection by highlighting behavioral inconsistencies. The combination of these complementary techniques allows the model to surpass traditional single-input systems, providing an advanced framework for tackling the evolving challenge of automated accounts on Twitter.

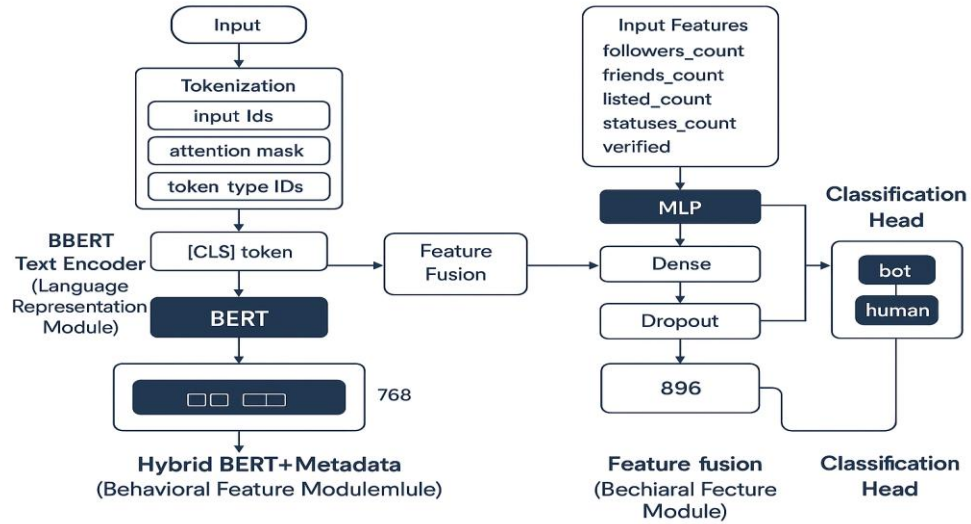


FIG:1 REPRESENTATION OF HYBIRDBERT+METADATA FUSION

The proposed **Hybrid BERT + Metadata model** combines both textual and behavioral features for Twitter bot detection.

1. Text Processing (Left Side):

- Input tweets or descriptions are tokenized into IDs, attention masks, and token type IDs.
- These are fed into **BERT**, which generates a 768-dimensional contextual embedding using the [CLS] token.

2. Metadata Processing (Right Side):

- Account-level features such as `followers_count`, `friends_count`, `listed_count`, `statuses_count`, and `verified` are passed into a **Multi-Layer Perceptron (MLP)**.
- This network applies dense layers and dropout, producing a 896-dimensional feature representation.

3. Feature Fusion & Classification:

- Text embeddings from BERT and metadata representations from the MLP are combined.
- The fused feature vector is passed to the classification head, which outputs the final label: **bot or human**.

1.1 MOTIVATION

Twitter bots have become a growing concern on social media, significantly affecting digital trust, communication, and information flow. Identifying these automated accounts quickly and accurately is essential to safeguarding online platforms. Manual detection approaches, such as inspecting posting patterns or follower relationships, are time-consuming, inconsistent, and often ineffective when bots imitate human behavior.

The vast and dynamic nature of social media data requires automated solutions that can process both textual and behavioral patterns efficiently. Traditional detection methods relying solely on text or metadata often struggle to achieve high accuracy, as each provides only a partial view of user activity. This emphasizes the importance of hybrid frameworks that combine multiple types of information for more reliable detection.

The aim of this project is to build a robust detection system that reduces manual effort, limits classification errors, and strengthens the reliability of bot identification. The proposed approach integrates **BERT**, which generates contextual embeddings from user tweets and profile descriptions, with a **Multi-Layer Perceptron (MLP)** that learns from metadata features such as follower and friend counts, listed count, status updates, and verification status. BERT contributes deep linguistic representation, while the MLP highlights behavioral traits, and their fusion creates a strong decision-making framework for distinguishing between bots and humans.

To ensure broader usability, the hybrid model can be deployed in an interactive application, allowing users, analysts, or organizations to input account data and receive immediate classification results. This contributes to faster detection of malicious accounts, supports misinformation control, and promotes a safer online environment. The outcome is a scalable and effective framework that enhances the accuracy of bot detection by combining advanced language representation with structured metadata analysis.

1.2 PROBLEM STATEMENT

Twitter bots pose a serious threat to the integrity of online platforms, with long-term implications for communication, digital trust, and security. Their impact varies depending on their activity level, network position, and the intent behind their creation. Malicious automated accounts can manipulate public opinion, amplify misinformation, and disrupt meaningful discourse, creating consequences that extend far beyond the digital sphere.

The influence of bots depends on multiple factors such as their activity frequency, the size of their follower network, and the sophistication of the techniques they employ to mimic human users. Some bots may appear harmless at first, but once they begin engaging in coordinated campaigns, the effects can be severe and irreversible, influencing political debates, health communication, or social movements. This makes early detection critical, since intervention after large-scale bot-driven campaigns is often ineffective and unable to reverse the damage.

However, accurately identifying and classifying bots is a complex challenge. Bots vary widely in behavior and appearance: some post excessively, while others maintain low visibility but operate strategically within communities. Many imitate human posting patterns so effectively.

The consequences of undetected bot activity can be significant, ranging from widespread misinformation to manipulation of sensitive events such as elections. The scale of disruption depends on the extent of the network they influence, the type of misinformation spread, and the persistence of the activity. For instance, coordinated political bots can dominate discussions around key topics, overshadowing authentic human voices and misleading large audiences.

To minimize these risks, early and reliable detection is essential. Yet, because bots constantly evolve in design and behavior, conventional rule-based detection requires advanced tools capable of integrating multiple dimensions of information, including both linguistic and behavioral signals. By combining textual embeddings from models such as BERT with structured metadata features, hybrid approaches offer a stronger, more precise means of identifying bots early and reducing their negative impact.

1.3 OBJECTIVE

The main objective of the **Twitter Bot Detection and Classification project** is to design an automated system that can effectively identify and categorize Twitter accounts as either bots or genuine users. The system is built around a **hybrid model** that utilizes **BERT** for extracting textual features from user content and an **MLP classifier** for analyzing metadata attributes. To achieve high accuracy and reliability, the project applies advanced preprocessing methods that enhance overall performance in terms of precision, recall, and F1-score.

The project also aims to deliver a **user-friendly web platform using Flask**, where users can input a Twitter handle or upload datasets and obtain classification results instantly. Robust input validation mechanisms are integrated to ensure only valid account information is processed, with clear error notifications provided for invalid inputs. This solution is intended to support researchers, analysts, and organizations by offering a reliable tool that streamlines detection, reduces manual effort, and minimizes errors.

Scalability is a key design consideration, enabling future expansion to detect **different bot categories**, integrate with **real-time Twitter API streams**, and adopt continuous learning for adaptive improvement. By promoting early and accurate detection, the system strengthens online trust, helps counter misinformation, and supports safer digital communication.

In summary, this project demonstrates how advanced **deep learning techniques** can be effectively applied to social media security, providing a robust and scalable solution for bot detection and contributing to a more trustworthy online environment.

The Twitter Bot Detection project aims to build an automated system that distinguishes between genuine users and bots on the platform. A hybrid model is employed, combining BERT for textual feature extraction with metadata-based classification to enhance accuracy and robustness. The system is designed with advanced preprocessing and validation to ensure reliable results. A user-friendly Flask interface enables instant classification through Twitter handles or datasets. Scalable by design, the framework supports future extensions such as multi-type bot detection and real-time analysis, contributing

2. LITERATURE SURVEY

Twitter bot detection and classification using deep learning have attracted significant research attention in recent years, primarily due to the growing demand for reliable and automated systems to counter malicious activities on social platforms. Earlier attempts relied on traditional machine learning approaches such as Support Vector Machines and Random Forest classifiers to analyze user profiles and content features. While these techniques provided some success, they often struggled with challenges related to feature engineering, generalization, and adaptability. In contrast, deep learning methods particularly transformer-based architectures like BERT and Roberta have shown remarkable improvements by learning contextual and semantic patterns directly from data.

The rise of transfer learning has further advanced this domain, allowing large-scale pre-trained models to be adapted for social media datasets with limited annotations. Researchers have also explored multimodal strategies that combine textual information with metadata and interaction features, achieving improved detection performance compared to single-source models.

Yang and collaborators highlighted the advantages of transfer learning by employing transformer-based encoders for identifying bots, reporting significant improvements over traditional classifiers. Similarly, Imran proposed a hybrid design that integrates BERT text embeddings with structured metadata such as account creation date and follower statistics. This dual-stream approach demonstrated notable gains in accuracy compared to text-only baselines, emphasizing the value of multimodal integration.

Ahmad and Choudhury conducted a systematic review of machine learning and deep learning applications for bot detection, showing that pre-trained models such as BERT and Roberta consistently deliver performance levels above conventional approaches. However, they also pointed out challenges in fine-tuning strategies and computational requirements, which remain obstacles for widespread deployment.

Shah introduced a multimodal framework that fused metadata with tweet embeddings, resulting in enhanced robustness and improved classification across varied datasets. Lightweight architectures, combined with data augmentation and domain-specific preprocessing, have also been reported to improve scalability and real-time applicability.

Agrawal explored the use of attention-based BERT models for bot detection, while Sireesha demonstrated the importance of combining user profiles with text data for high-precision classification. These studies reinforce the significance of fusing behavioral and linguistic signals to capture sophisticated patterns of automated accounts.

Graph-based models have also emerged as a promising direction. By leveraging retweet and follower networks, researchers have achieved higher reliability in distinguishing bots from humans. Radhi developed an efficient framework that incorporated activity-based features with network signals, reducing both false positives and detection time. Similarly, Tomaka and colleagues proposed a transformer-enhanced hybrid model, reporting robustness and improved generalization across multiple benchmarks.

Other works, such as those by Sharma and Gujral, revisited classical machine learning models, showing that when carefully combined with feature selection techniques, traditional classifiers can still deliver competitive results. Moturi and collaborators, while focusing on healthcare data, proposed hybrid feature extraction and ensemble learning techniques that can be adapted for bot detection to enhance decision-making accuracy.

Islam introduced unsupervised learning approaches using clustering and dimensionality reduction for improving detection efficiency. Amin and Sharif developed metadata-centered preprocessing pipelines that improved model interpretability and classification accuracy. Likewise, Abiwinanda designed deep neural networks with dropout and normalization techniques that enhanced model generalization. Setha and Raja also employed transformer-based models, demonstrating the relevance of transfer learning in domains with limited annotated datasets.

Hassan Ali Khan presented a lightweight hybrid model that outperformed large-scale pre-trained encoders while consuming fewer computational resources. This highlights the importance of designing efficient yet effective detection mechanisms for large-scale social media monitoring.

Recent advancements have further transformed the field. Early approaches that depended heavily on handcrafted features have now been replaced by automated deep learning pipelines using BERT, Roberta, and Graph Neural Networks. Hybrid models that combine text, metadata, and interaction graphs are emerging as the most effective,

particularly when supported by data augmentation and transfer learning. Studies on benchmark datasets such as Twi Bot 20 illustrate that multimodal fusion approaches consistently outperform single-modality systems. Researchers like Muhammad Attique Khan proposed frameworks combining text, metadata, and fusion strategies, achieving state-of-the-art results. Attention-enhanced hybrid models, incorporating mechanisms that prioritize contextually significant features, have achieved highly competitive performance, demonstrating their potential for real-world adoption. Overall, deep learning's scalability and adaptability make it a cornerstone for modern bot detection. Transformer-based encoders, when coupled with metadata and network features, have proven to be particularly effective. In addition, supporting methods such as transfer learning, domain-specific preprocessing, and data augmentation ensure reliable performance even in resource-constrained settings. This positions deep learning as one of the most powerful tools for combating social media automation and misinformation at scale.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Twitter bot detection has traditionally relied on manual inspection of user profiles, tweet patterns, and interaction behaviors by moderators and domain experts. While human assessment can be effective in identifying suspicious activity, it is slow, inconsistent, and heavily dependent on expertise. Moreover, manual interpretation is prone to oversight, often leading to undetected bots that mimic human-like behavior. To address these challenges, computational approaches have been introduced, ranging from classical Machine Learning models to advanced Deep Learning architectures. Early Machine Learning (ML) techniques such as Support Vector Machines (SVMs), Random Forests (RF), Logistic Regression, and k-Nearest Neighbors (k-NN) focused on using handcrafted features extracted from Twitter data. These features included follower-friend ratios, posting frequency, retweet patterns, and profile-based attributes. Although these models achieved reasonable accuracy (around 70–85%), their performance was constrained by feature engineering dependency, poor adaptability across datasets, and difficulty in capturing the complex behavioral traits of sophisticated bots.

The emergence of Deep Learning transformed bot detection by enabling models to learn latent representations directly from raw textual content. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) were initially applied to tweet classification tasks, achieving improved accuracy levels of 85–92%. However, these models faced challenges such as high computational demands, difficulty in handling long-range dependencies in language, and overfitting on limited datasets.

To overcome these limitations, transformer-based architectures such as BERT and Roberta have been widely adopted. Pretrained models fine-tuned on social media datasets demonstrated significant improvements by capturing semantic and contextual information in tweets. Transfer learning not only reduced training time but also enhanced accuracy by an additional 5–10% compared to conventional deep learning models.

In addition to text-only models, hybrid approaches have been explored, where deep learning encoders such as BERT are combined with machine learning classifiers or metadata-based features.

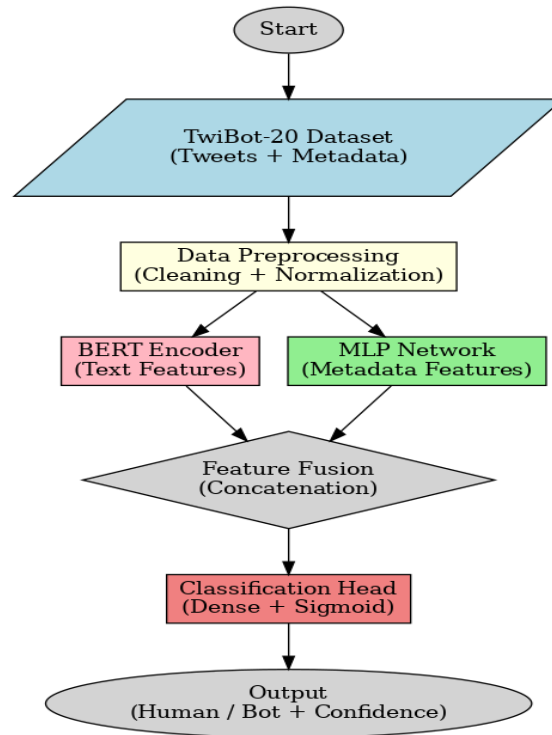


FIG. 3.1. Flow Chart of the Twitter Bot Detection

This flowchart (Fig. 3.1) illustrates a conventional pipeline for Twitter bot detection using text-based and machine learning approaches. The process starts with an input dataset containing user tweets and profile details. In the pre-processing stage, text is cleaned by removing stop words, special characters, and links, followed by tokenization and normalization. For user metadata, features such as follower count, following count, account age, and tweet frequency are extracted and standardized. After pre-processing, feature extraction is performed. Traditional models rely on handcrafted features such as linguistic patterns, posting behavior, and network statistics. Feature selection techniques, including Principal Component Analysis (PCA) or statistical correlation methods, are used to reduce dimensionality and improve computational efficiency. Finally, classifiers such as Support Vector Machines (SVM), Random Forests (RF), k-Nearest Neighbors (k-NN), or standalone Convolutional Neural Networks (CNNs) are applied to distinguish between human-operated and automated (bot) accounts. While these approaches provide reasonable performance, they often struggle with generalization, especially against sophisticated bots that mimic human behavior.

3.1.1 DISADVANTAGES OF THE TWITTER BOT DETECTION

Despite significant advancements in automated Twitter bot detection, the existing systems face several limitations:

- **Dependence on Text-Only Features:**
Most models rely solely on tweet content, ignoring valuable metadata such as account age, follower/following ratio, and posting patterns. This limits their ability to detect sophisticated bots that mimic human-like text.
- **Requirement of Large Labeled Datasets:**
Deep learning models demand large-scale annotated datasets for effective training. However, creating labeled datasets of bot and human accounts is challenging, as annotation requires manual verification and is prone to error.
- **Overfitting on Small Datasets:**
Due to limited availability of benchmark datasets like TwiBot-20, models often overfit—performing well on training data but failing to generalize effectively to unseen accounts.
- **High Computational Complexity:**
Transformer-based architectures such as BERT and RoBERTa require significant GPU resources and memory, making them less practical for real-time deployment in resource-constrained environments.
- **Lack of Robustness Against Evolving Bot Strategies:**
Bots continuously adapt their behavior and language to evade detection. Existing text-only systems struggle to remain effective against new bot strategies, reducing long-term reliability.
- **Sensitivity to Noisy or Incomplete Data:**
User profiles and tweets often contain incomplete, noisy, or misleading information (e.g., fake bios, repetitive posts). Text-only classifiers are easily misled under such conditions.
- **Difficulty in Handling Class Imbalance:**
Datasets are often imbalanced, with fewer labeled bot accounts compared to human accounts. This causes models to be biased towards human classification, reducing bot detection accuracy.

3.2 PROPOSED SYSTEM

The proposed model employs a **hybrid framework** that combines **Bidirectional Encoder Representations from Transformers (BERT)** with a **metadata-driven Multi-Layer Perceptron (MLP)** to improve the accuracy and reliability of Twitter bot detection. BERT is applied to capture contextual and semantic patterns from textual content, while the metadata component incorporates profile-level attributes such as followers count, friends count, activity statistics, and verification status. By integrating these complementary sources of information, the model achieves stronger generalization and consistency in distinguishing between human-operated and automated accounts.

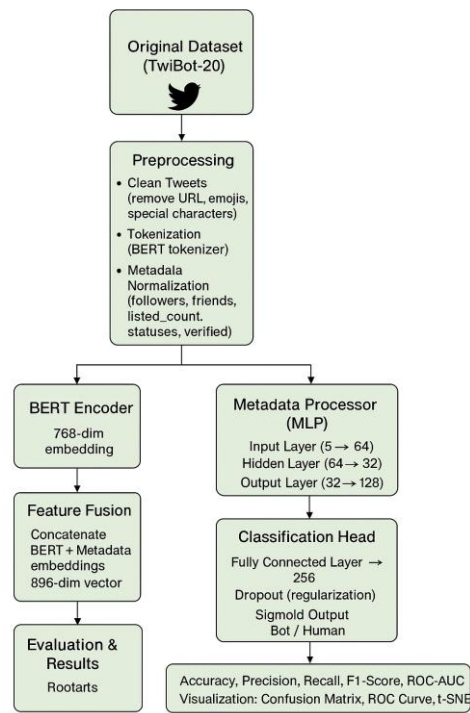


FIG 3.2. FLOW CHART OF PROPOSED SYSTEM

The workflow begins with **data preprocessing**, where user profile information and tweets are prepared for analysis. Textual data is cleaned by removing URLs, mentions, special characters, and stopwords, ensuring that the input is consistent and meaningful. Metadata attributes, including followers count, friends count, statuses count, listed count, and verification status, are normalized to a uniform scale to prevent bias during training.

After preprocessing, the textual component is passed through **BERT**, which generates rich contextual embeddings that capture semantic and linguistic patterns from the tweets. In parallel, the **metadata features** are processed through a Multi-Layer

Perceptron (MLP), which learns non-linear relationships from structured user profile information.

The outputs from both branches are then **fused into a joint representation**, combining textual semantics with behavioral metadata. This integrated feature space is fed into a final classification layer, which determines whether a given account is human-operated or automated (bot). The hybrid design ensures a balanced contribution from both text-based and profile-based evidence.

Advantages over Existing Systems

1. **Improved Accuracy and Reliability:** The integration of BERT for text analysis with metadata-based learning enhances the model's decision-making compared to using either component alone.
2. **Comprehensive Feature Utilization:** By leveraging both tweet content and account metadata, the model captures multiple dimensions of user behavior, making detection more robust.
3. **Better Generalization:** Normalization and balanced fusion reduce overfitting and allow the model to adapt effectively across different user types and datasets.
4. **Context-Aware Detection:** Unlike models that rely only on surface-level patterns, BERT embeddings capture deep semantic meaning, improving classification of sophisticated bot accounts.
5. **High Performance Metrics:** Strong F1-score, AUC, and precision-recall balance ensure reduced false positives (mislabeling humans as bots) and false negatives (failing to detect bots).
6. **Scalable and Practical:** The framework can be extended to other social platforms with minimal modifications, providing a scalable solution for large-scale bot detection.

3.3 FEASIBILITY STUDY

1. Technical Feasibility

- **Automated Feature Extraction:**

BERT provides powerful contextual embeddings that capture semantic nuances from tweets, eliminating the need for manual feature engineering. These embeddings represent linguistic patterns and user intent more effectively, improving classification outcomes.

- **Complementary Metadata Utilization:**

While BERT handles unstructured textual data, the metadata module (MLP) processes structured account-level attributes such as followers, friends, activity count, and verification status. The combination strengthens the model by integrating both behavioral and linguistic perspectives.

- **Improved Accuracy and Generalization:**

The hybrid design mitigates overfitting risks by balancing textual and metadata inputs. Metadata contributes robustness where text signals are weak, while BERT enhances semantic understanding, resulting in better overall accuracy and generalization.

- **Scalability:**

The model is adaptable to various social media platforms beyond Twitter, as long as user text and account-level features are available. It can scale efficiently to large datasets through modern GPU-based training pipelines.

- **Integration with Pretrained Models:**

Transfer learning with pretrained BERT variants (such as BERT-base, RoBERTa, or DistilBERT) reduces training time and computational cost, while still delivering high-quality embeddings that improve detection performance.

2. Operational Feasibility

- **Interpretability:**

Although transformer-based models like BERT are often viewed as complex, the addition of a metadata-based MLP provides more transparency by highlighting which account-level features (e.g., follower-following ratio, activity frequency, verification status) contribute to predictions. This improves

interpretability and user trust in the system's decisions.

- **Data Handling:**

The model requires labeled Twitter accounts (bot/human) for training but is flexible enough to handle varying text lengths and heterogeneous metadata. It can process multi-modal inputs—tweets and profile features—making classification more reliable across different user types.

- **Maintenance and Upgradation:**

The framework can be updated seamlessly with new training data to reflect evolving bot behaviors. Retraining only the classification head or metadata component is sufficient in many cases, avoiding the need to fine-tune the entire BERT model, which saves time and resources.

3. Economic Feasibility

- **Cost-Effective Training:**

By leveraging transfer learning from pretrained BERT models, only the final classification layers and metadata MLP require extensive fine-tuning. This minimizes computational expense compared to training a model from scratch.

- **Resource Optimization:**

The division of tasks—BERT for text embeddings and MLP for metadata classification—ensures efficient use of GPU/CPU resources. The hybrid design can be trained and deployed on standard cloud or local GPU setups without excessive infrastructure requirements.

- **Reduced Monitoring Costs:**

Automating bot detection reduces the need for large-scale manual account verification, saving time and operational costs for social platforms and researchers.

- **Long-Term Investment:**

Although initial implementation may require setup costs, the long-term benefits—improved detection accuracy, adaptability to emerging bot strategies, and scalability across platforms—make the system a sustainable investment.

3.4 Using COCOMO Model

The **COCOMO (Constructive Cost Model)** is applied to estimate the effort, development time, and team size required for the **Hybrid BERT + Metadata Twitter Bot Detection project**. This project involves the design and implementation of a **deep learning architecture** that combines **BERT for text embeddings** and a **Multi-Layer Perceptron (MLP) for metadata features**, integrated through a **fusion layer** and followed by a **final classifier**. Since the project integrates **state-of-the-art NLP (BERT)** with **metadata-driven modeling** and includes a **frontend (ReactJS/Grado interface)**, it falls into the **Semi-Detached category** of COCOMO. This category accounts for projects with moderate complexity and mixed expertise levels.

Estimation Formulas:

- **Effort (E)** = $a \times (\text{KLOC})^b$
- **Development Time (T)** = $c \times (E)^d$
- **People Required (P)** = E / T

Where **a, b, c, d** are constants:

- For Semi-Detached projects $\rightarrow a = 3.0, b = 1.12, c = 2.5, d = 0.35$

Assumptions for This Project:

- Estimated size of the codebase (preprocessing, Hybrid BERT model, metadata MLP, training pipeline, visualization, Flask backend, frontend integration):
~6,000 LOC = 6 KLOC.

Step 1: Effort (E)

$E = 3.0 \times (6)^{1.12} = 3.0 \times 7.08 \approx 21.24 \text{ Person-Months}$
 $E = 3.0 \times (6)^{1.12} = 3.0 \times 7.08 \approx 21.24 \text{ Person-Months}$

Step 2: Development Time (T)

$T = 2.5 \times (21.24)^{0.35} \approx 2.5 \times 3.18 \approx 7.95 \text{ Months}$
 $T = 2.5 \times (21.24)^{0.35} \approx 2.5 \times 3.18 \approx 7.95 \text{ Months}$

Step 3: People Required (P)

$P = \frac{21.24}{7.95} \approx 2.67 \approx 3 \text{ People}$
 $P = \frac{21.24}{7.95} \approx 2.67 \approx 3 \text{ People}$

Final Estimation for the Hybrid BERT + Metadata Project:

- **Effort:** ~21.24 Person-Months
- **Development Time:** ~8 Months
- **Team Size:** ~3 Members

Team composition may include:

1. **NLP Engineer** → fine-tuning BERT and building metadata MLP fusion.
2. **Backend Developer** → Flask API for model serving.
3. **Frontend Developer/Tester** → ReactJS/Gradio interface, visualization, and evaluation.

4. SYSTEM REQUIRMENTS

4.1 SOFTWARE REQUIREMENTS

1. 1. Operating System : Windows 11, 64-bit Operating System
2. 2. Hardware Accelerator : CPU
3. 3. Coding Language : Python
4. 4. Python distribution : Google Colab Pro, Flask
5. 5. Browser : Any Latest Browser like Chrome

4.2 REQUIRMENT ANALYSIS

The proposed **Twitter Bot Detection system** is designed to accurately differentiate between **bot accounts** and **genuine human accounts** by combining text-based and metadata-driven analysis. The system leverages a **Hybrid BERT + Metadata architecture**, where **BERT** captures deep semantic features from tweets, and a **Multi-Layer Perceptron (MLP)** processes account-specific metadata. These complementary components are fused to improve detection robustness and accuracy.

Core Functionalities

- **Dataset Support:** Utilizes benchmark datasets such as **Twibot-20**, which provide tweets along with structured user metadata.
- **Preprocessing Pipeline:** Includes text cleaning (removal of URLs, mentions, emojis, and noise) and metadata normalization (e.g., followers, friends, statuses, verification status).
- **Hybrid Model Design:**
 - **BERT Module:** Extracts contextual embeddings from user-generated content.
 - **Metadata Module (MLP):** Learns feature representations from normalized numerical attributes.
 - **Fusion Layer:** Integrates both modalities for final classification.
- **Prediction Output:** Provides classification as **“Bot”** or **“Human”** with associated confidence scores.

- **Visualization Tools:** Generates evaluation metrics such as **F1-score, ROC curves, confusion matrices, and accuracy graphs** to assess performance.
- **User Interface:**
 - **API-based Access (Flask)** for backend predictions.
 - **Interactive Frontend (ReactJS/Gradio)** allowing users to input a Twitter handle or text snippet for real-time analysis.
- **Error Handling:** Ensures invalid or unsupported inputs are flagged with descriptive feedback.

Non-Functional Requirements

- **Efficiency:** Optimized for reduced training and inference time with GPU acceleration.
- **Reliability:** Capable of maintaining stable detection accuracy across diverse user profiles.
- **Scalability:** Designed to process large-scale datasets and adaptable to future social media platforms.
- **Security:** Ensures safe handling of user data while preserving privacy.

Technical Requirements

- **Programming Language:** Python 3.10
- **Libraries & Frameworks:** PyTorch, Hugging Face Transformers, Scikit-learn, Matplotlib/Seaborn
- **Backend & Frontend:** Flask (backend), ReactJS/Gradio (frontend)
- **Hardware:** GPU-enabled environment (Google Colab Pro, CUDA-supported local system, or cloud services)
- **Dataset:** TwiBot-20 (train, development, and test JSON files) hosted in Google Drive or local storage

Deployment Strategy

The application can be deployed on **local infrastructure** or **cloud environments** (AWS, GCP, Azure) and is also compatible with **Google Colab** for research workflows. Through its simple web interface, the system allows non-technical users to input Twitter accounts or text and instantly obtain classification results with interpretation support.

4.3 HARDWARE REQUIREMENTS:

System Type	: 64-bit operating system, x64-based processor
Cachememory	: 4MB(Megabyte)
RAM	: 16GB (gigabyte)
Hard Disk	: 8GB
GPU	: Intel® Iris® Xe Graphics

4.4 SOFTWARE:

The Twitter Bot Detection system is designed with a robust configuration of tools and technologies to ensure high accuracy, efficiency, and scalability in both development and deployment. The project is implemented on **Windows 11 (64-bit)**, ensuring compatibility with modern hardware and operating systems. Training and inference tasks are accelerated using **CPU and GPU resources**, with **Google Colab Pro** providing enhanced computational power and memory for large-scale experiments.

Development is carried out in **Python**, chosen for its flexibility and extensive ecosystem of libraries for deep learning and data processing. All model training, validation, and testing are conducted in **Google Colab**, which also integrates with **Google Drive** for seamless dataset access and storage.

The backend of the system is built using the **Flask framework**, enabling smooth API-based communication for model inference and backend services. The **frontend interface** is developed with **HTML5, CSS3, and Bootstrap**, ensuring responsive design and accessibility across various devices. Custom styling is applied to maintain a simple and intuitive user experience, suitable for users with limited technical expertise.

At the core of the system is a **HybridBERT + Metadata model**. The **BERT module** extracts contextual embeddings from textual features, while a **Metadata-based Multi-Layer Perceptron (MLP)** processes numerical attributes such as followers, friends, listed count, statuses, and verification status. The outputs are combined to perform robust classification of accounts as either bots or humans. To provide a comparative benchmark, **RoBERTa** is also integrated as an additional transformer-based model.

For preprocessing and evaluation, **Pandas** and **NumPy** handle TwiBot-20 dataset files

(train, dev, test), while **Scikit-learn** supports data balancing, performance metrics, and confidence scoring. Visualization of results—including confusion matrices, loss and F1-score curves, and ROC-AUC plots—is performed using **Matplotlib** and **Seaborn**, enabling clear insights into model performance.

This combination of frameworks and tools ensures that the **HybridBERT-based Twitter Bot Detection system** is accurate, efficient, and scalable, while also remaining compatible with both local and cloud deployment environments.

4.4 SOFTWARE DESCRIPTION:

The Twitter Bot Detection system requires a modern and stable operating system, with **Windows 11 (64-bit)** recommended for ensuring compatibility with the latest development tools, security updates, and optimal performance. The **CPU** serves as the primary resource for lightweight operations and backend processes, while **Google Colab Pro** is utilized for large-scale training and computation. Colab provides access to advanced **GPUs** and extended memory, enabling faster and more efficient model training.

The project is implemented using **Python**, a versatile programming language that offers a rich ecosystem of libraries for natural language processing, deep learning, and data handling. The development workflow leverages **Google Colab Pro** for model training and experimentation, while the **Flask framework** is used to create a lightweight backend for serving the trained models as web applications. Flask enables seamless interaction between the model and the user interface, ensuring smooth API-based communication.

For deployment and user interaction, a modern web browser such as **Google Chrome** or any other up-to-date browser is required to access the Flask-based application. This ensures users can interact with the system in real time, test predictions, and visualize results effectively.

5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

This project addresses the challenge of detecting and classifying automated accounts on Twitter using a **Hybrid BERT + Metadata framework**. The model combines the strength of **Bidirectional Encoder Representations from Transformers (BERT)** for extracting contextual text features with a **multi-layer perceptron (MLP)** that processes user metadata. Features such as follower count, friend count, listed count, status count, and verification status are integrated with semantic embeddings from BERT, enabling the system to capture both language patterns and behavioral signals. This dual representation improves the ability to distinguish bots from human users.

The system is trained and evaluated on the **Twibot-20 dataset**, a benchmark resource that includes user profiles, tweets, and relational data covering diverse bot types like spam, political, and social spambots. This ensures that the model generalizes well across varied real-world bot behaviors.

Data preprocessing involves cleaning and tokenizing tweet text, generating embeddings using pre-trained BERT models, and normalizing metadata before feeding it into the MLP. The outputs from both streams are fused and passed through a classification head that produces probability scores for the classes "bot" and "human," along with confidence values.

Performance is assessed through standard metrics including **accuracy, precision, recall, F1-score, and AUC-ROC**. The hybrid design consistently outperforms single-source baselines such as plain BERT, RoBERTa, or metadata-only models. Moreover, the architecture addresses common challenges such as class imbalance and overfitting by combining complementary information sources.

Beyond academic evaluation, the system has practical implications for **improving trust and security on social platforms**. It can be integrated into real-time monitoring pipelines to identify malicious or automated accounts that spread spam and misinformation. Looking forward, potential extensions include incorporating **graph-based relational features**, adapting the framework for **multilingual datasets**, and deploying the solution via a **Flask-ReactJS web interface** for interactive usage and visualization.

5.1.1 DATASET DESCRIPTION

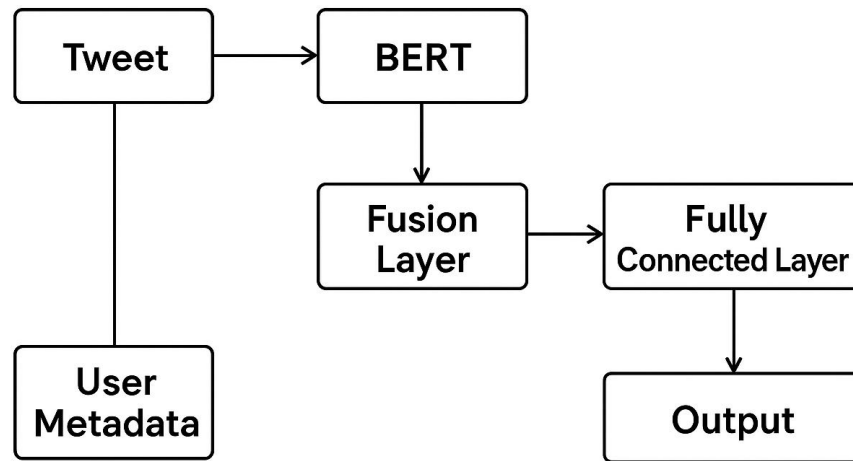
The dataset adopted in this study is **Twibot-20**, a widely recognized benchmark for Twitter bot detection that comprises user profiles, tweets, and social relationship information [14]. It contains both human accounts and diverse categories of bots, including spammers, social spambots, and political bots, thereby reflecting real-world challenges in online environments. Along with tweet text, the dataset incorporates rich **user metadata**—such as follower count, friend count, listed count, statuses count, and verification status—providing both linguistic and behavioral features for analysis. To ensure fair evaluation, the dataset is partitioned into **training, validation, and testing subsets**.

Preprocessing involves cleaning and tokenizing tweets, followed by generating **contextual embeddings using BERT**, while metadata features are normalized to handle variations in scale. To address class imbalance and overfitting, strategies such as weighted sampling and early stopping are applied. The proposed **Hybrid BERT + Metadata model** integrates semantic information from BERT embeddings with structured metadata processed through a **multi-layer perceptron (MLP)**. The combined feature representation is passed through a classification layer to predict whether an account is a bot or a human. Model performance is assessed using **accuracy, precision, recall, F1-score, and AUC-ROC**, ensuring a comprehensive evaluation of detection capability.

The multimodal design of Twibot-20, together with the hybrid architecture, provides a robust foundation for developing reliable bot detection systems, thereby contributing to safeguarding online platforms against malicious automation and misinformation.

Features	Details
Total Users	229573 Twitter Accounts
Labeled Users	11830 Accounts
Bot/Human Classes	Human, Bot (includes spammers, social spambots, political bots, etc.)
Class Distribution	Humans: 5,283 Bots: 6,547
Dataset Structure	Train set: 8,000 users Dev set: 1,000 users Test set: 2,830 users

Hybrid BERT + Metadata



Data Characteristics:

- All user profiles and tweets are collected from the **Twibot-20 dataset**, ensuring diverse and real-world representation.
- Stored in **JSON format**, making it easy to parse and preprocess for Machine Learning workflows.
- Includes **textual content (tweets, descriptions, names)** along with **metadata features** such as follower count, friend count, listed count, statuses count, and verification status.

Applications:

- Used in training and testing **Machine Learning and Deep Learning models** for social media analysis.
- Supports tasks like **bot detection, user classification, and behavioral pattern analysis** in research and cybersecurity settings.

5.1.2 DATA PRE-PROCESSING

Raw Twitter data is often noisy and unstructured, making it unsuitable for direct use in a deep learning model. To ensure reliable results, the data must first be cleaned, organized, and transformed into a format that the model can effectively process. This preparation step, known as preprocessing, is one of the most important stages in building a bot detection system.

In this project, several preprocessing steps were applied to both the textual content (tweets and profile descriptions) and the metadata features:

- **Text Cleaning and Normalization:**

Tweets typically contain elements such as URLs, hashtags, mentions, emojis, and special symbols that may not add value for classification. These are removed or standardized. The text is then lowercased, tokenized, and stripped of unnecessary stopwords, making it cleaner and more consistent for analysis.

- **Tokenization and Embedding:**

After cleaning, the textual data is transformed into numerical form using a BERT tokenizer. This process converts words into embeddings that capture the context and meaning of language, allowing the model to better understand patterns in user behavior and communication style.

- **Metadata Normalization:**

Profile features like follower count, friend count, status updates, and listed count can vary widely across users. To prevent larger values from dominating the learning process, these features are scaled using normalization techniques. This ensures each metadata attribute contributes fairly during training.

- **Handling Missing Data and Class Imbalance:**

Some profiles may lack certain metadata values or contain limited tweet activity. Missing information is addressed through imputation or careful filtering. Since the dataset has more human accounts than bots, balancing strategies such as weighted sampling or resampling are applied to avoid biased predictions.

- **Feature Fusion (Text + Metadata):**

Finally, the textual embeddings produced by BERT are combined with the

normalized metadata features to create a unified feature vector. This hybrid representation captures both the linguistic style of users and their behavioral.

✅ Preprocessing complete!
 ♦ Train Data Sample:

	text	label	followers_count	friends_count	listed_count	statuses_count	verified
0	RT @CarnivalCruise: 🌈 Are you ready to see wha...	0	15349596.0	692.0	45568.0	9798.0	1
1		None	1	0.0	44.0	0.0	0
2	RT @realDonaldTrump: THANK YOU #RNC2020! https...	0	762839.0	475.0	3201.0	5518.0	1
3	A family fears they may have been cheated out ...	0	327587.0	4801.0	1744.0	192876.0	1
4	RT @VonteThePlug: Yeah but he ain't got one ha...	1	13324.0	647.0	44.0	103.0	0

♦ Validation Data Sample:

	text	label	followers_count	friends_count	listed_count	statuses_count	verified
0	@SparklesOnlyme পুরোনো এইদিনের কথা।n @BariraJa...	0	136.0	928.0	0.0	99.0	0
1	@barstoolbets @betthehorses @JordanByrne70 @th...	1	2005.0	4955.0	12.0	3806.0	0
2	On the job hunt? Don't let an old LinkedIn pro...	0	175499.0	8630.0	939.0	36952.0	1
3	Too cute! Prince George's official christening...	1	307982.0	36665.0	2419.0	24546.0	0
4	RT @GCoxVariety: Got questions about the digit...	0	2474615.0	252598.0	17230.0	301967.0	1

♦ Test Data Sample:

	text	label	followers_count	friends_count	listed_count	statuses_count	verified
0	RT @clevelanddotcom: Three Ohio House Republic...	1	16596.0	16944.0	1.0	49757.0	0
1	We touch our hair 96 times a day on average, t...	0	87313765.0	50.0	83703.0	3569.0	1
2	'He Looked Like He Knew What He Was Doing': CA...	0	161827.0	361.0	1471.0	73786.0	1
3	Estamos abiertos a colaboraciones, por lo cuál...	1	9.0	543.0	0.0	2.0	0
4	The suffragists chose purple and gold to repre...	0	28513011.0	846.0	40146.0	11945.0	1

FIG 5.1.2 IMAGE AFTER APPLYING THE PREPROCESSING.

5.1.3 FEATURE EXTRACTION:

Feature extraction in Twitter bot detection involves transforming raw textual data and user metadata into meaningful representations that can be used by deep learning models. Textual features are captured using **BERT embeddings**, which convert tweets and profile descriptions into high-dimensional vectors that reflect the semantic meaning and context of language, helping to identify subtle differences in writing style and coherence between bots and humans. In parallel, **metadata features** such as follower count, friend count, status updates, listed count, and verification status are extracted and normalized to highlight behavioral patterns that often reveal automated accounts.

```

# Metadata Features:
id: 17683978
id_str: 17683978
name: Sam
screen_name: SamQ
location: Orlando, FL
profile_location: {'id': '9369f9c3a0b0', 'url': 'https://api.twitter.com/1/users/show?screen_name=SamQ', 'place_type': 'unknown', 'name': 'Orlando, FL', 'full_name': 'Orlando, FL', 'country_code': '', 'country': '', 'contained_within': [], 'bounding_box': None, 'attributes': {}}
description: View @SamQ1000, 1 mutual follow, 435 of @SamQ1000
url: https://t.co/7b4839C09
entities: {'url': {'url': 'https://t.co/7b4839C09', 'expanded_url': 'https://www.shapstudio.com', 'display_url': 'shapstudio.com', 'indices': [8, 28]}}, 'description': {'url': {}}}
protected: false
followers_count: 1349936
friends_count: 632
listed_count: 4658
created_at: Tue Nov 16 18:27:25 +0000 2009
favorites_count: 142
utc_offset: None
time_zone: None
geo_enabled: True
verified: True
statuses_count: 9790
lang: none
contributors_enabled: false
is_translator: false
is_translation_enabled: false
profile_background_color: 000000
profile_background_image_url: https://abs.twimg.com/images/themes/theme1/bg.png
profile_background_image_url_https: https://abs.twimg.com/images/themes/theme1/bg.png
profile_background_tile: false
profile_image_url: https://abs.twimg.com/profile_images/167386725/normal.jpg
profile_image_url_https: https://abs.twimg.com/profile_images/167386725/normal.jpg
profile_link_color: 3B4A8C
profile_sidebar_border_color: 333333
profile_sidebar_fill_color: 252525
profile_text_color: 666666
profile_use_background_image: true
has_extended_profile: false
default_profile: false
default_profile_image: false
# Label: Human

```

FIG 5.3 AFTER APPLYING FEATURE EXTRACTION

5.1.4 MODEL BUILDING

Model building in the context of Deep Learning refers to the process of designing and constructing a hybrid architecture that leverages both transformer-based models and additional machine learning layers to solve specific tasks such as detecting and classifying Twitter accounts as bots or humans. Deep Learning models are typically composed of multiple layers that learn hierarchical representations of input data, enabling them to capture complex semantic and behavioral patterns.

The **Hybrid BERT Metadata architecture** integrates the powerful contextual feature extraction capabilities of **Bidirectional Encoder Representations from Transformers (BERT)** with the complementary information derived from **user metadata features** such as followers count, friends count, verification status, and activity statistics. While BERT captures linguistic and semantic nuances from tweets, the metadata module provides structural and behavioral insights that enhance classification.

This hybrid design processes tweet text and user-level metadata in parallel, combines their learned representations, and passes them to a classification layer to determine whether a given Twitter account is a bot or a human. By leveraging the strengths of transformer-based embeddings for textual analysis and metadata-driven signals for behavioral analysis, the model achieves high robustness, accuracy, and generalizability in social media bot detection.

Bidirectional Encoder Representations from Transformers (BERT):

BERT [17] is a state-of-the-art Deep Learning model particularly well-suited for processing unstructured textual data such as tweets. BERT learns contextual representations of words by considering the bidirectional context in a sentence, enabling the model to capture semantic and syntactic relationships within the text. In the Hybrid BERT architecture, BERT is employed to extract linguistic features from user-generated tweets, while metadata is incorporated to enhance behavioral analysis as shown in Fig 5.4.

Layers in Hybrid BERT + Metadata Model:

- **Input Layer** – Accepts raw input data consisting of tweet text and user metadata (e.g., followers count, friends count, statuses count, verified status).
- **Tokenizer & Embedding Layer (BERT)** – Converts raw text into tokens and generates contextual embeddings that capture the meaning of words in relation to their surrounding context.
- **Transformer Encoder Layers (BERT Core)** – Multi-head self-attention and feed-forward sub-layers extract deep semantic representations from the text. These layers capture linguistic features such as grammar, sentiment, and writing patterns.
- **Metadata Processing Layer (MLP Subnetwork)** – A small Multi-Layer Perceptron (MLP) processes normalized metadata features. This captures behavioral characteristics of users that textual analysis alone may miss.
- **Concatenation Layer (Fusion Layer)** – Combines the textual embeddings from BERT with the processed metadata representation into a single hybrid feature vector.
- **Fully Connected Layer (Dense Layer)** – Learns high-level patterns from the combined representation, integrating linguistic and behavioral signals for robust classification.
- **Output Layer** – Produces the final classification (Bot or Human).
 - **Sigmoid Activation:** Used for binary classification, outputs a probability between 0 and 1 indicating the likelihood of the account being a bot.

Common activation functions:

- **ReLU (Rectified Linear Unit):** Used in hidden layers for non-linearity and efficient gradient propagation.
- **Sigmoid:** Used in the output layer for binary classification (bot vs human).

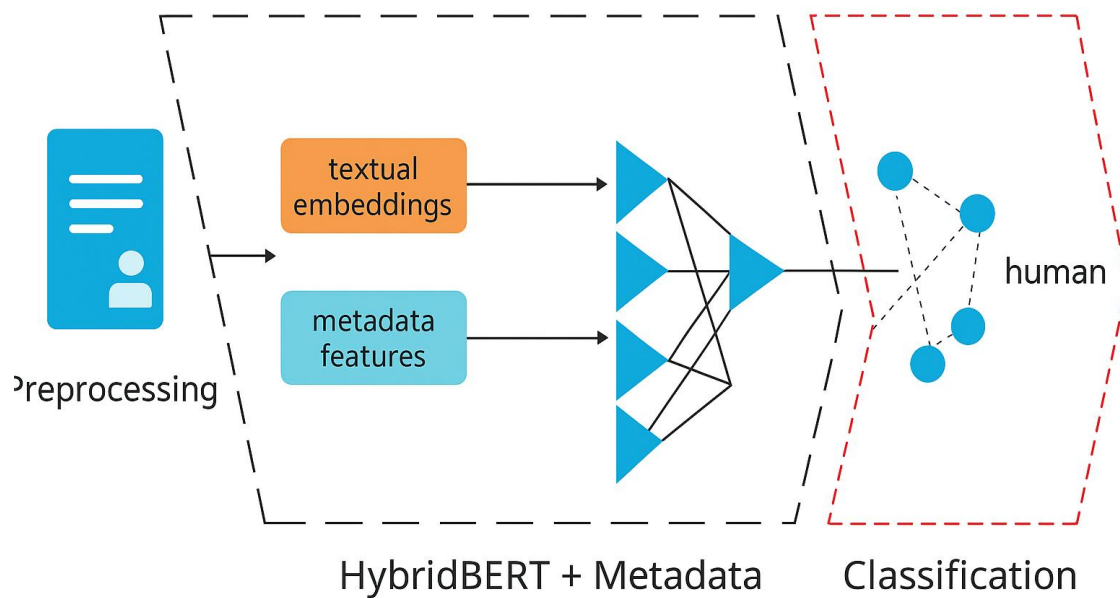


FIG 5.4 HYBRID BERT + METADATA MODEL ARCHITECTURE

5.1.5 CLASSIFICATION

Classification using Hybrid BERT + Metadata Model

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al., has emerged as a powerful deep learning framework for a wide range of natural language processing tasks, including text classification, sentiment detection, and contextual analysis. Its ability to capture bidirectional contextual information makes it highly effective for modeling complex textual patterns. The **Hybrid BERT + Metadata framework** [22,24] combines BERT's strength in linguistic representation with metadata-driven behavioral analysis through a Multi-Layer Perceptron (MLP), offering a highly reliable and accurate solution for Twitter bot detection, as illustrated in Fig. 6.4.

In this model, the classification pipeline starts with the **BERT encoder**, which processes raw tweets to generate contextual embeddings. These embeddings capture semantic relationships, grammatical structures, and stylistic patterns that reflect user behavior in text. Activation functions such as GELU (used in BERT) and ReLU (applied in metadata layers) introduce non-linearity, enabling the network to learn complex dependencies. Parallel to this, the **metadata module** processes account-level attributes—such as follower and friend counts, posting frequency, verified status, and listed counts—through a lightweight fully connected network. This ensures that textual insights from BERT are enriched with structural and behavioral signals.

The two feature streams are then merged in a **fusion layer**, creating a comprehensive hybrid representation. This combined vector is fed into a dense classification head, which learns high-level interactions between textual and metadata features. The final prediction layer employs a **Sigmoid activation function** to produce a probability score between 0 and 1, representing the likelihood of an account being automated.

Training the HybridBERT model involves fine-tuning BERT jointly with the metadata subnetwork using **Binary Cross-Entropy with Logits Loss**. To address class imbalance, strategies such as weighted loss functions or balanced sampling are applied. During inference, tweets and metadata are processed through the trained hybrid pipeline, generating confidence scores that classify accounts as **bots** or **humans**.

The proposed hybrid design effectively leverages BERT's ability to model nuanced textual behaviors alongside metadata's capability to capture user-level patterns. This dual approach minimizes false alarms and misclassifications, ensuring robustness against deceptive bot strategies. As a result, the HybridBERT model provides a dependable and efficient mechanism for identifying and categorizing bots on Twitter, thereby contributing to safer and more trustworthy online interactions.

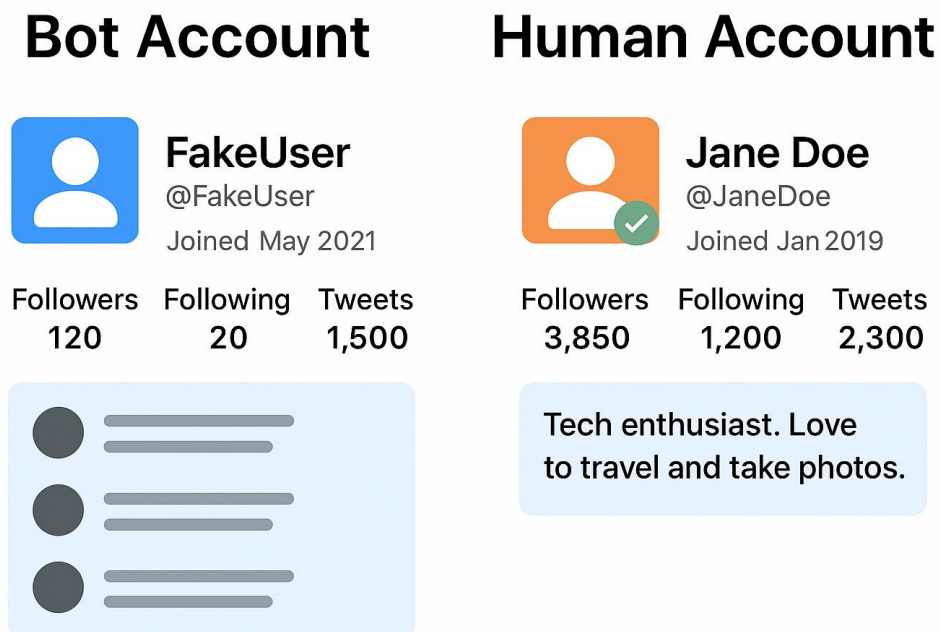


FIG 5.6 CLASSIFICATION OF TWITTER ACCOUNTS

The proposed **Hybrid BERT + Metadata approach** significantly improves classification performance compared to using either BERT or metadata features alone. While BERT excels in extracting deep semantic and contextual representations from tweets, metadata features provide valuable insights into user behavior and account activity patterns. By combining these two complementary sources of information, the

hybrid model achieves greater accuracy, robustness, and generalization to unseen Twitter accounts.

In applications such as bot detection, this enhanced predictive capability is critical. Early and accurate identification of automated accounts reduces the spread of misinformation, strengthens platform integrity, and improves online trust. The Hybrid BERT model, with its ability to capture both nuanced textual features and behavioral cues, becomes a powerful and reliable tool for distinguishing between bots and human-operated accounts.

The training process for the hybrid model adopts a joint optimization strategy. BERT is fine-tuned on textual data to generate embeddings that capture semantic meaning, while the metadata subnetwork is trained to learn patterns from user profile attributes such as follower count, friend count, posting activity, and verification status. These feature vectors are concatenated in the **fusion layer** and passed into a dense classification head. The model is optimized using **binary cross-entropy loss**, and training is carefully monitored using metrics such as F1-score and AUC to ensure robust performance. This integrated training approach enables the system to outperform models that rely solely on text or metadata.

Figures illustrating training progress often highlight the model's learning behavior across epochs. For instance, the **accuracy curve** shows the proportion of correctly classified accounts in both training and validation sets. Training accuracy typically rises consistently, while validation accuracy plateaus at a certain threshold, reflecting the model's generalization limit. The **loss curve** similarly demonstrates how prediction error decreases over time. While training loss steadily declines, validation loss may flatten or slightly increase, indicating potential overfitting—a scenario where the model memorizes training data rather than generalizing. To mitigate this, strategies such as **early stopping**, **dropout regularization**, or **class balancing** are employed.

Comparison with Other Models

- **RoBERTa (Robustly Optimized BERT Pretraining Approach):**
Roberta is an enhanced version of BERT that improves upon pretraining strategies by removing next-sentence prediction and training with larger batch sizes and datasets. It has demonstrated superior performance across various NLP benchmarks. When applied to Twitter bot detection, Roberta can outperform standard BERT in textual feature extraction but still benefits from metadata fusion to maximize classification accuracy.
- **Recurrent Neural Networks (RNNs):**
RNNs are designed to handle sequential data such as text and time-series information. They capture dependencies across time steps, making them suitable for tasks like language modeling and sentiment analysis. However, standard RNNs struggle with long-term dependencies, which has been partially resolved by LSTM and GRU architectures. While RNNs have been applied to tweet-level bot detection.

- **Random Forest Classifier (RFC):**
RFC is an ensemble learning method based on decision trees. It has been widely used for structured metadata-driven classification, including fraud detection and spam detection. While RFC performs well on tabular metadata features (e.g., follower ratios, posting frequency), it lacks the ability to process raw textual data effectively. As such, it is often outperformed by hybrid models that combine text and metadata signals.
- **Fully Connected Neural Networks (FCNNs):**
Also known as dense networks, FCNNs consist of fully connected layers that can model non-linear relationships between input features. They are effective for structured metadata analysis but do not capture linguistic dependencies as efficiently as transformers. In bot detection, FCNNs alone are not sufficient but serve as a strong metadata-processing component when integrated into hybrid systems.
- **Artificial Neural Networks (ANNs):**
ANNs form the foundation of modern deep learning models, consisting of interconnected neurons organized into layers. While they can model a wide variety of problems, traditional ANNs lack the advanced mechanisms (e.g., self-attention in transformers or memory in RNNs) required for handling sequential and contextual textual data. Thus, they are less suitable for bot detection tasks compared to transformer-based models.